

Theoretical Cosmology

From fields to the Universe

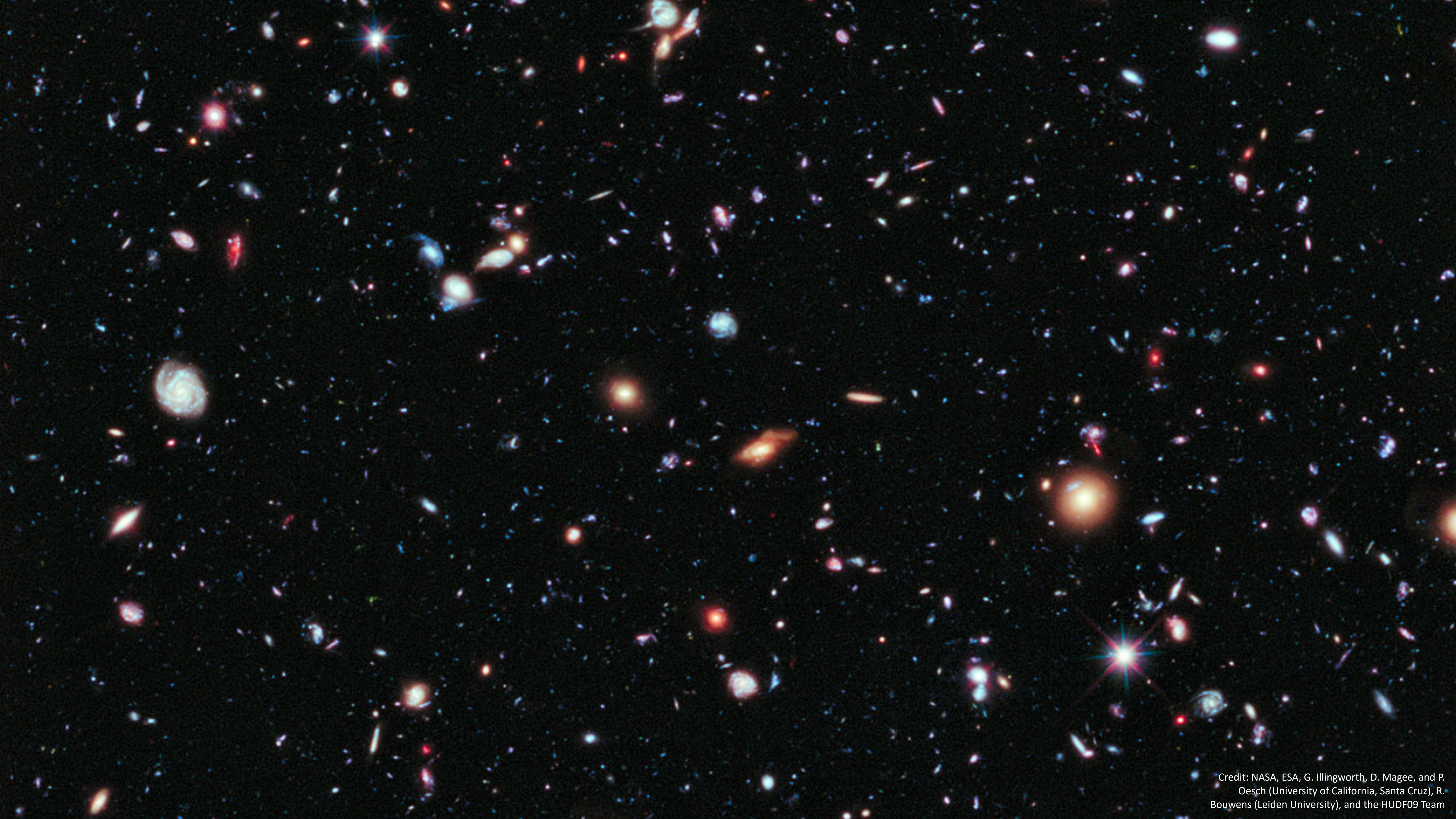
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27 & 28 July 2026 | VSOA-10, ICISE, Vietnam



Credit: NASA, ESA, G. Illingworth, D. Magee, and P.
Oesch (University of California, Santa Cruz), R.
Bouwens (Leiden University), and the HUDF09 Team

Lecture plan

L1: Fields and The Universe

L2: Essentials of cosmological perturbations

L3: Theories of gravity

Handout 1

1. Find equations of motion for the following systems. What is their number of degrees of freedom ?

$$L = \frac{1}{2}\dot{q}^2 - \frac{1}{2}k^2q^2$$

$$L = \dot{q}^2$$

$$\mathcal{L} = \frac{1}{2}(\partial^\mu A_\mu)^2$$

2. Give an example of a homogeneous but anisotropic, and isotropic but inhomogeneous spacetime.

3. Consider spatially flat and closed Universes, which are filled with dust. Find how its energy density and scale factor change in time for each of them?

4. What is the condition that the Universe accelerates? Assume that the Universe is filled with some constant energy density, with $p = -\varepsilon_\Lambda$. What is the scale factor as a function of time in this case?

5. Show that the energy-momentum tensor of a scalar field can be written in terms of the perfect fluid. Can it imitate a cosmological constant?